Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study

Panlin Li[a], Qingli Zhang[a,b], Qitai Jiao[c], Jiamei Sun[a] and Hua Huang[a,b,*]

[a] *College of Mathematics and Physics, Xinjiang Agricultural University, Urumqi 830052, Xinjiang, China*
[b] *Research Center for Numerical Simulation and Data Analysis, Xinjiang Agricultural University, Urumqi 830052, Xinjiang, China*
[c] *College of Mechanical and Electrical Engineering, Xinjiang Agricultural University, Urumqi 830052, Xinjiang, China*

ARTICLE INFO

Keywords:
Apple
Near-infrared spectroscopy
Declared-origin discrimination
Few-shot learning
Meta-learning
Benchmark
Domain shift

ABSTRACT

This paper contributes a benchmark and a leakage-controlled evaluation protocol for cross-year near-infrared (NIR) origin discrimination of commercial apples. The benchmark is a nine-domain dataset covering three harvest years (2018, 2019, 2025) and four origins (Xinjiang, Shandong, Shanxi, Gansu), with 2,424 fruits, 4,529 spectra and 229 bands over the 590–1001 nm overlap window. We define a fold-restricted leave-one-year-out (LOEO-Year) protocol with three components: (i) a unified single-face query rule applied to all methods on every fold; (ii) a strict source-only z-score preprocessing variant that is fully target-blind; and (iii) a shared-encoder versus fold-restricted-encoder split that quantifies evaluation leakage. A shared encoder inflated the three-fold macro F_1 at $K = 5$ by +0.266 (0.931 vs. 0.665) over the best fold-restricted configuration; the gap widened to +0.333 on the LOEO-2025 fold, which confounds year, measurement-protocol and class-space change. Fifty-repetition experiments over $K \in \{1, 3, 5, 10, 20, 50\}$ compared four few-shot baselines under the unified protocol—vanilla Prototypical Networks (ProtoNet), RBF-SVM, partial least squares discriminant analysis (PLS-DA) and nearest centroid—plus an exploratory ProtoNet-C source-centroid-fusion ablation that did not outperform vanilla ProtoNet on any fold. In the three-fold mean, vanilla ProtoNet leads for $K \leq 10$ and SVM overtakes for $K \geq 20$; the crossover is sensitive to source-domain composition. We also report 1D CNN and 1D ResNet supervised closed-set sanity checks. A 2×2 ablation on the 2025 fold decomposed the face-mean gain into a query-side contribution of +0.055 and a support-side contribution of +0.011 in F_1, informing in-field scanning. Future LOEO studies should declare target-domain exposure and report a strict source-only baseline.

1. Introduction

The apple (*Malus domestica* Borkh.) is one of the most widely consumed temperate fruits in the world, and Chinese production accounts for more than 47 % of the global supply. Geographic origin is an important attribute for both pricing and food-safety traceability, since price differentials between protected origins can be substantial and false labelling has well-documented economic and regulatory consequences. Conventional analytical approaches—stable-isotope-ratio analysis, multi-element fingerprinting and sensory evaluation—offer high specificity but require specialised laboratory infrastructure and detection cycles of several hours, which makes them unsuitable for batch-scale screening (Danezis, Tsagkaris, Camin, Brusic and Georgiou, 2016; Gonzalvez, Armenta and de la Guardia, 2009).

Near-infrared (NIR) diffuse-reflectance spectroscopy is an attractive alternative: the measurement is reagent-free, takes less than a minute per scan, and yields directly interpretable spectra (Nicolai, Beullens, Bobelyn, Peirs, Saeys, Theron and Lammertyn, 2007; Blanco and Villarroya, 2002; Walsh, Blasco, Zude-Sasse and Sun, 2020). Spectral variation in apple fruit reflects the joint contribution of soluble solids, organic acids, polyphenols and tissue water content, all of which are influenced by soil geochemistry, climate and orchard management; the resulting high-dimensional patterns can provide an origin-related spectral fingerprint for apples under commercial conditions (Cen and He, 2007; Bureau, Cozzolino and Clark, 2019), in line with broader evidence that NIR can encode geographic fingerprints across food matrices (Woodcock, Downey, Kelly and O'Donnell, 2007). For commercial apples sourced from wholesale markets, NIR spectra additionally encode varietal differences, storage conditions, ripeness and supply-chain handling. Accordingly, we use "origin traceability" to mean origin-label discrimination under commercial-batch conditions. Partial least squares discriminant analysis (PLS-DA) is the long-established baseline for this task and typically achieves high accuracy under single-year, single-instrument conditions (Geladi and Kowalski, 1986; Wold, Sjöström and Eriksson, 2001; Westad and Marini, 2015). With larger training sets, support vector

*Corresponding author. Email: huanghua@xjau.edu.cn.
huanghua@xjau.edu.cn (H. Huang)
ORCID(s): 0000-0003-0615-9402 (H. Huang)

machines (SVMs) and convolutional networks have outperformed PLS-DA in several reported settings (Cortes and Vapnik, 1995; Ekramirad, Khaled, Doyle, Loeb, Donohue, Villanueva and Adedeji, 2022; LeCun, Bengio and Hinton, 2015).

A persistent gap is that single-year laboratory performance does not transfer reliably to subsequent harvest seasons. NIR models calibrated in a controlled laboratory commonly exceed 90 % same-year accuracy, yet the train–test distribution gap caused by year-to-year changes in climate, soil nutrient status and instrument drift can substantially degrade cross-year performance (Feudale, Woody, Tan, Myles, Brown and Ferre, 2002). Dedicated cross-year studies for apple NIR remain scarce, with cross-year performance typically reported only as a secondary check. Existing calibration-transfer methods—direct standardisation (DS), piecewise direct standardisation (PDS) and gradient-boosted recalibration—were primarily designed for instrument-to-instrument transfer and require a batch of labelled target samples on the deployment side, a prerequisite that is often unavailable at the start of a new season (Fearn, 2001).

Few-shot learning (FSL) is a plausible alternative in this setting. By accumulating cross-task inductive bias, FSL methods generalise rapidly from a small number of labelled samples (Wang, Yao, Kwok and Ni, 2020; Snell, Swersky and Zemel, 2017). Prototypical networks (Snell et al., 2017) are well suited to this problem: each class is represented by the mean embedding of its support set, and classification reduces to nearest-prototype distance. Meta-learning has been applied to hyperspectral remote-sensing classification (Liu, Yu, Yu, Zhang, Wan and Wang, 2019), but its application to food NIR remains limited. Spatial context, which assists discrimination in remote-sensing imagery, is absent for point-spectrum food NIR, and this difference has not been adequately addressed.

Three methodological gaps remain in the literature. First, no public apple NIR dataset spans multiple harvest years together with a standardised leave-one-year-out evaluation protocol; cross-year results from different teams are therefore not directly comparable. Second, when a shared encoder is pre-trained on the full data, including the held-out year, before evaluation, the resulting protocol leaks target-year information into the representation, and the magnitude of this leakage has not been formally quantified for episodic meta-learning on NIR data. Third, multi-position averaging is known to denoise apple spectra, but the contribution of averaging on the support side (prototype estimation) and on the query side (prediction stability) has not been disentangled.

The contribution of this paper is a benchmark and a leakage-controlled evaluation protocol, not a new classifier. We do not propose a new few-shot algorithm. Instead, we (i) deposit a public nine-domain apple NIR benchmark; (ii) define a fold-restricted LOEO-Year protocol with a strict source-only pre-processing variant and a unified single-face query rule; (iii) quantify evaluation leakage under shared-encoder versus fold-restricted training; (iv) report a wide baseline panel under the same protocol, including modern 1D deep baselines that have so far been missing from cross-year apple NIR studies. The exploratory source-centroid-fusion variant ProtoNet-C is reported only as an ablation around the fold-restricted ProtoNet (its best setting is $\alpha = 0$, which collapses to vanilla ProtoNet), and the multi-face mean is reported as a separate ablation rather than as part of the headline numbers. The four contributions in detail are as follows.

Nine-domain cross-year benchmark dataset. We describe the first apple NIR benchmark dataset covering three harvest years (2018, 2019, 2025) and four origins (Xinjiang, Shandong, Shanxi, Gansu), comprising nine independent collection domains, 4,529 spectra and 229 bands, together with a standardised fold-restricted LOEO-Year evaluation protocol, deterministic seeds, fruit-identifier split tables, and full instrument and acquisition specifications. The data are internal to Xinjiang Agricultural University and are not publicly released; they are available from the corresponding author on request (the Data availability statement).

Quantification of evaluation leakage and a strict source-only pre-processing baseline. Through a paired comparison between shared-encoder and fold-restricted-encoder protocols, we show on this benchmark that the shared encoder inflates the LOEO macro F_1 at $K = 5$ by +0.266 over the best fold-restricted baseline (0.931 vs. 0.665), and the gap widens to +0.333 on the 2025 fold. We further introduce a strict source-only z-score pre-processing variant whose statistics are computed only on source-fold spectra (Section 3.5), making the protocol fully target-blind, and report the residual gap against the weakly-transductive year-wise variant. To our knowledge, this is the first systematic quantification of evaluation leakage within an episodic meta-learning framework on NIR data.

Unified single-face baseline panel under one protocol. Over $K \in \{1, 3, 5, 10, 20, 50\}$, we conduct 50-repetition experiments comparing four few-shot baselines—nearest centroid, PLS-DA, RBF-SVM and vanilla ProtoNet—under fold-restricted encoders and a unified single-face query rule. ProtoNet-C is reported only in the controlled α-ablation (Section 4.6); $\alpha = 0$ is best on every fold, collapsing to vanilla ProtoNet. As separate supervised closed-set sanity checks we report a 1D CNN and a 1D ResNet trained on every face spectrum of the source fold (Section 4.8); these are not framed as a supervised, full-source extrapolation of the few-shot panel because their closed-set softmax has no mechanism for an unseen origin class on LOEO-2025. All numbers in the headline panel are produced with the unified single-face query rule on every fold; multi-face averaging on the 2025 fold is reported only in the dedicated face-mean ablation. The performance crossover between vanilla ProtoNet and SVM sits between $K = 10$ and $K = 20$ in the three-fold mean (Section 4.10 shows it is not stable to source-domain composition), delineating a method-selection boundary indexed by the available support size.

Asymmetric decomposition of face-mean gain (separate ablation). A 2×2 ablation on the multi-face 2025 fold decomposes the face-mean gain into a query-side contribution of +0.055 in F_1 and a support-side contribution of +0.011, providing practical guidance for in-field scanning protocols. The face-mean ablation is reported separately and does not enter the headline cross-method table.

2. Related Work

2.1. NIR spectroscopy for fruit origin traceability

NIR spectroscopy has accumulated more than two decades of application to fruit quality and authentication. Early work on apple confirmed that diffuse reflectance captures sufficient compositional variation—soluble sugars, organic acids and mineral-bound water—for PLS-DA to discriminate origins (Cen and He, 2007; Nicolai et al., 2007; Luo, Huan, Fu, Wen, Cheng, Zhou, Wu, Shen and Yu, 2011), and broader food-matrix studies verified the feasibility of NIR-based geographic fingerprinting (Woodcock et al., 2007). Subsequent apple and adjacent-commodity studies achieved discrimination accuracies above 90 % under controlled single-year conditions (Walsh et al., 2020; Eisenstecken, Stürz, Robatscher, Lozano, Zanella and Oberhuber, 2019; Li, Huang, Zhao and Zhang, 2013; Ma, Xia, Inagaki and Tsuchikawa, 2018), with a relatively stable pre-processing consensus around SNV (Barnes, Dhanoa and Lister, 1989), Savitzky–Golay derivatives (Savitzky and Golay, 1964) and multiplicative scatter correction (Rinnan, van den Berg and Engelsen, 2009; Mishra, Roger, Marini, Biancolillo and Rutledge, 2021). With enough training data, SVM and random forests can match or exceed PLS-DA (Cortes and Vapnik, 1995; Breiman, 2001; Ekramirad et al., 2022), and convolutional and attention-based neural networks (He, Zhang, Ren and Sun, 2016; Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017) have started to appear in food NIR applications (Mishra and Passos, 2021; Benmouna, Pourdarbani, Sabzi, Fernández-Beltrán, García-Mateos and Molina-Martínez, 2022). Most apple NIR origin studies, however, are single-year; cross-year transfer is rarely a primary objective.

2.2. Cross-year domain shift, calibration transfer and meta-learning

Calibration transfer addresses adapting chemometric models across instruments or conditions (Feudale et al., 2002), with direct/piecewise direct standardisation (Fearn, 2001) as a long-established tool. Liu, Zhang, Jiang and Liu (2025), Li, Li, Jiang and Liu (2022) and Pan, Wu, Hu, Li, Zhang and Zhao (2024) extended these ideas to apple firmness, origin and seasonal robustness, but all assume labelled target samples are available. Domain-adaptation theory (Ben-David, Blitzer, Crammer, Kulesza, Pereira and Vaughan, 2010; Pan and Yang, 2010; Ganin, Ustinova, Ajakan, Germain, Larochelle, Laviolette, Marchand and Lempitsky, 2016; Long, Cao, Wang and Jordan, 2015; Peng, Bai, Xia, Huang, Saenko and Wang, 2019) has been widely deployed in computer vision but is harder to translate to small-sample food NIR. Few-shot meta-learning (Hospedales, Antoniou, Micaelli and Storkey, 2022; Wang et al., 2020) provides an alternative: prototypical networks (Snell et al., 2017) build class centres from K-sample mean embeddings; matching networks (Vinyals, Blundell, Lillicrap, Wierstra and Kavukcuoglu, 2016), relation networks (Sung, Yang, Zhang, Xiang, Torr and Hospedales, 2018) and MAML (Finn, Abbeel and Levine, 2017) extend the comparison and adaptation paradigms. Re-evaluations (Tian, Wang, Krishnan, Tenenbaum and Isola, 2020; Chen, Liu, Kira, Wang and Huang, 2019) show that a well-trained embedding backbone with nearest-centroid inference can match many sophisticated meta-learners. Hyperspectral remote sensing (Liu et al., 2019) has adopted these methods, exploiting spatial context that is unavailable for point-spectrum food NIR. Mishra and Passos (2021) applied transfer learning to food traceability but used no episodic evaluation. The evaluation-contamination problem—distribution leakage caused by overlap between encoder pre-training data and the LOEO test fold—has not been formally identified or quantified in any prior NIR study.

3. Materials and Methods

3.1. Dataset construction

Apple NIR diffuse-reflectance spectra were collected over three harvest years (2018, 2019, 2025) from four major Chinese production regions: Xinjiang (XJ), Shandong (SD), Shanxi (SX) and Gansu (GS). All samples were predominantly Fuji and were purchased from local wholesale markets, with sampling distributed across multiple orchard batches per origin. *Origin labels are wholesale-market-declared values rather than chain-of-custody-certified labels*; the benchmark therefore evaluates origin-label discrimination under commercial-batch conditions, not detection of a verified geochemical signal (limitation discussed in Section 5). Samples were stored at room temperature (20–25 °C) for at most three days before measurement. Samples from different years come from their respective harvest seasons, so year-to-year ripeness and sugar differences are expected sources of cross-year domain shift. The 2018 and 2019 datasets contain three origins each (XJ, SD, SX); 2025 has three origins (XJ, SD, GS), making GS a novel-class challenge for the 2025 fold. The nine collection domains are numbered E1–E9 in year-by-origin order. The 2018/2019 measurements used a fibre-optic Vis–NIR point spectrometer (590–1250 nm, 2,048 native bands, CCD-based grating spectrometer with integrating-sphere reflectance probe and as-shipped firmware; integration time 8 ms, 32 scans averaged per acquisition; Spectralon white-reference and dark-current re-acquired every 30 min); one spectrum per fruit at the most uniform equatorial face. The 2025 measurements used a portable Vis–NIR push-broom hyperspectral imaging system (391–1001 nm, 346 native bands, silicon CMOS, $f/2.8$ telecentric lens, halogen line illumination through a 50/50 beam-splitter, factory-default firmware; integration time 12 ms, 5 line-scans averaged per pixel; Spectralon reference co-imaged with each fruit, dark frame at batch start); five face positions per fruit (top, bottom, three lateral faces 120° apart), each spectrum extracted as the mean of a manually drawn region of interest on uniform skin. Cross-year harmonisation cropped both instruments to the 590–1001 nm common overlap window, applied a 15-point second-order Savitzky–Golay smoother, and resampled 2018/2019 spectra to the 2025 grid by cubic-spline interpolation, yielding 229 uniformly distributed bands. Calibration certificates, dark/white-reference logs, and full acquisition tables are held alongside the spectra at Xinjiang Agricultural University and are available from the corresponding author on request (the Data availability statement).

There is an important asymmetry across years: 2018 and 2019 use a single equatorial face per fruit (single-face), so the number of spectra equals the number of fruits; 2025 uses four to five faces per fruit (multi-face). The three years together contain 2,424 fruits and 4,529 spectra (Table 1). When the distinction between fruit and spectrum matters, we use the words explicitly. The multi-face design is exploited in the face-mean ablation (Section 3.6).

Pre-processing follows a three-stage pipeline (Fig. 1) designed to preserve year-to-year distribution differences, which is essential for evaluating cross-year generalisation. (i) Savitzky–Golay smoothing with a 15-point window and a second-order polynomial (Savitzky and Golay, 1964) is applied with fixed filter coefficients that depend only on the window length and not on the data. (ii) An SNV transform (Barnes et al., 1989) is applied to each spectrum to remove additive and multiplicative scattering effects; SNV is purely per-spectrum and does not use any data-derived statistics across spectra. (iii) Optionally, a year-wise z-score normaliser can be applied as an extra stage; we explicitly study three variants—no z-score (default), year-wise z-score using each year's own statistics (weakly transductive), and a strict source-only z-score where the mean and standard deviation are estimated only on the source-fold spectra and applied identically to source and target—and quantify the effect of the choice in Section 3.5. The default pipeline does not apply a year-wise z-score, so all reported main-table results are produced by SG smoothing followed by SNV alone, which is supervised- and target-blind. Joint cross-year z-score standardisation, which would erase year-level covariate shift and inflate LOEO performance, is explicitly avoided.

Table 1
Dataset statistics: number of fruits and spectra by year and origin. The full 3-year × 4-origin grid contains 12 theoretical domain cells, of which 9 are filled in practice (three origins are sampled per year; XJ/SD/SX in 2018/2019, XJ/SD/GS in 2025). Empty cells: Gansu is a new origin that appears only in 2025 and was not collected in 2018/2019, which makes Gansu a novel-class challenge in the 2025 fold.

Year	Origin	Dom.	Fruits	Scans	Spectra
2018	Xinjiang (XJ)	E1	327	1	327
2018	Shandong (SD)	E2	298	1	298
2018	Shanxi (SX)	E3	270	1	270
2019	Xinjiang (XJ)	E4	341	1	341
2019	Shandong (SD)	E5	286	1	286
2019	Shanxi (SX)	E6	253	1	253
2025	Xinjiang (XJ)	E7	204	4–5	∼864
2025	Shandong (SD)	E8	215	4–5	∼910
2025	Gansu (GS)	E9	230	4–5	∼980
Total	—	E1–E9	2,424	—	4,529

What the default pre-processing pipeline does and does not use. SNV is per-spectrum and label-free, and uses no statistics aggregated across spectra; the Savitzky–Golay coefficients are fixed and data-independent. The default pipeline (SG + SNV only) is therefore both supervised- and unsupervised-target-blind: it uses no aggregated spectral statistics from the target year and no target-year labels. Section 3.5 additionally contrasts this default with two z-score variants—weakly transductive year-wise z-score and strict source-only z-score—to give a quantitative reading of how much room the year-wise distributional moment can move LOEO performance.

The final feature space consists of 229 spectral variables per sample, covering the visible-NIR window in which the third and higher overtones of O–H and C–H stretching vibrations and chlorophyll/carotenoid electronic transitions are observed; this window contains information on soluble solids, organic acids and water content, all of which have been associated with origin discrimination in apple (Nicolai et al., 2007; Pissard, Fernández Pierna, Baeten, Sinnaeve, Lognay, Mouteau, Dupont, Rondia and Lateur, 2013; Lammertyn, Peirs, De Baerdemaeker and Nicolai, 2000).

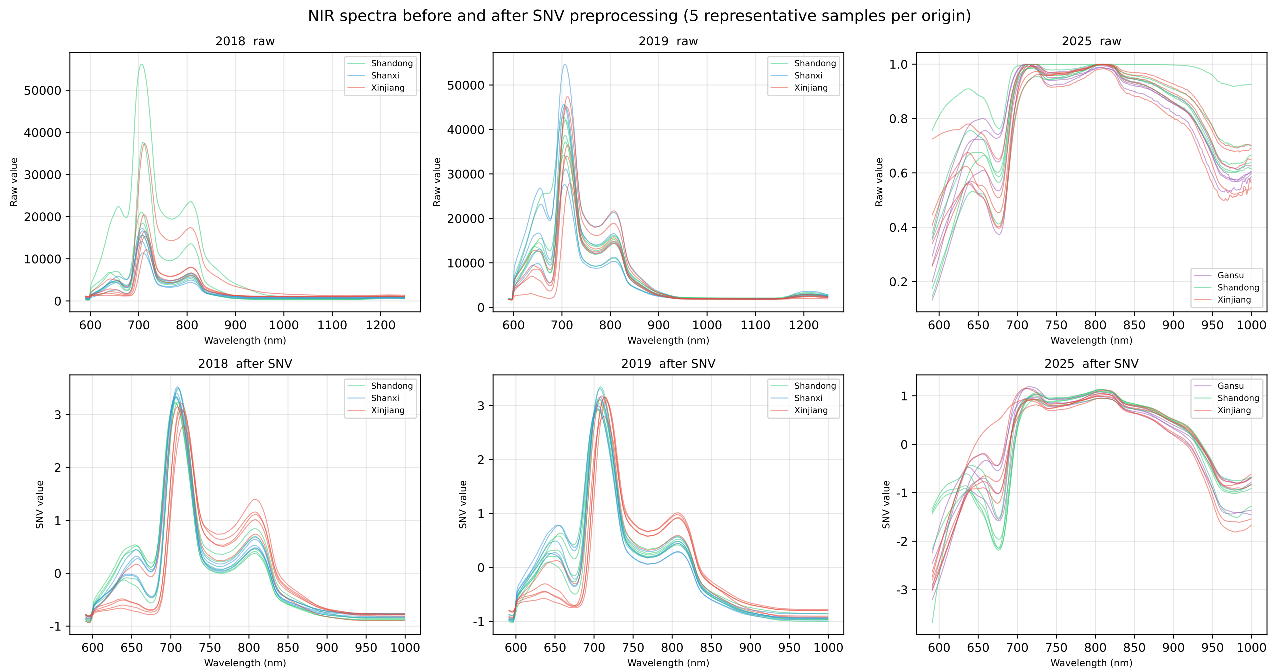

Figure 1: NIR spectra of three years × three origins, before and after pre-processing. Five randomly chosen representative samples are shown per origin. Left column: raw reflectance spectra; right column: spectra after the SNV transform.

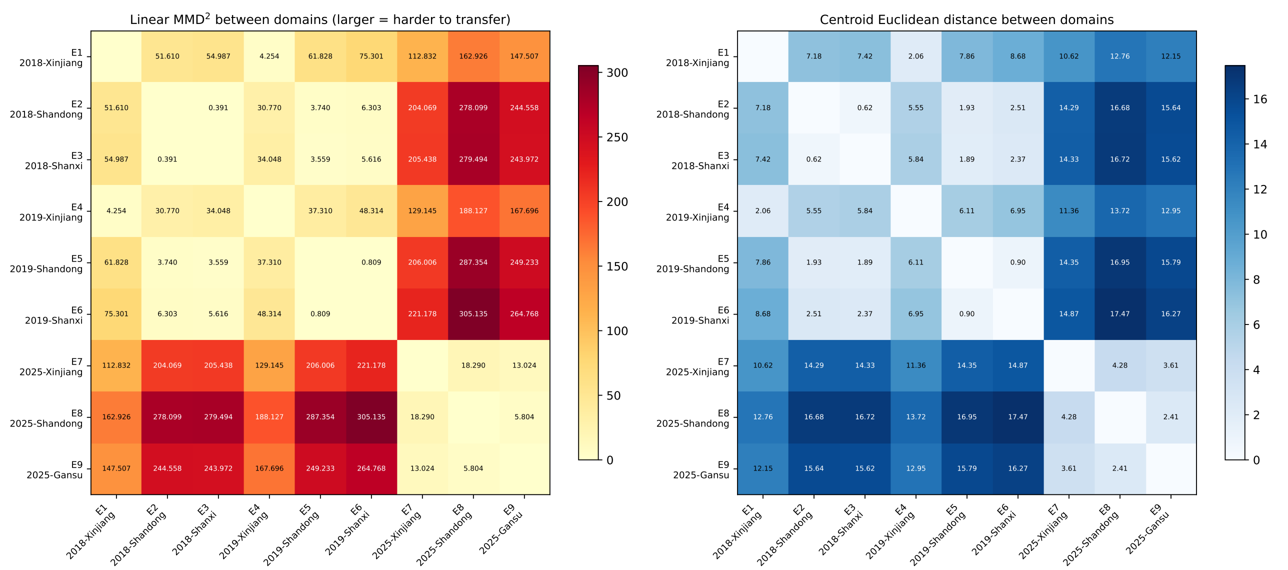

Figure 2: Distribution-difference matrices across the nine collection domains. Left: linear MMD^2; right: centre-point Euclidean distance. Darker colours indicate larger inter-domain differences.

Figure 2 further provides two distribution-distance measures between the nine collection domains. Whether measured by the linear MMD^2 or by the centre-point Euclidean distance, the three 2025 domains are systematically more distant from the six 2018/2019 single-face domains than the latter are from each other, indicating that the difficulty of cross-year transfer comes not only from class change but also from the joint drift of measurement protocol and year-specific conditions.

3.2. Evaluation framework: leave-one-year-out (LOEO-Year)

The primary evaluation framework is **leave-one-year-out** (LOEO-Year; in this paper we keep the abbreviation LOEO to refer to this year-level held-out protocol, distinguishing it from the generic "leave-one-out" cross-validation). Each harvest year is in turn designated as the target domain, while the remaining two years serve as the source domains. This defines three evaluation folds: (i) LOEO-2018: source = {2019, 2025}, target = 2018; (ii) LOEO-2019: source = {2018, 2025}, target = 2019; (iii) LOEO-2025: source = {2018, 2019}, target = 2025. The 2025 fold constitutes the most difficult evaluation scenario, because Gansu is a new origin not present in the source, while Xinjiang and Shandong appear in the source under different measurement conditions (single-face vs. multi-face). All reported numbers are means over 50 independent repetitions per fold, each repetition drawing new episode samples and a new random seed. The primary performance metric is macro F1, which weights all origin classes equally regardless of any class imbalance.

Taxonomy of evaluation protocols. We distinguish three protocol classes: (i) *fold-restricted*—encoder trained only on source-domain episodes; (ii) *unsupervised target exposure*—encoder pre-trained on all data including unlabelled target spectra (semi-/self-supervised); (iii) *labelled target adaptation*—encoder/classifier fine-tuned on labelled target samples. We use "evaluation contamination" to refer to the distribution leakage of (ii) when used as an LOEO baseline. The default pipeline (Section 3.1, SG + SNV) is fully supervised- and unsupervised-target-blind; the optional year-wise z-score adds a weakly-transductive moment, and a strict source-only z-score variant is reported for protocol-pedantic comparisons (Section 4.7). All results in this paper use the fold-restricted encoder unless explicitly noted otherwise.

Multiple confounders in LOEO-2025. LOEO-2025 confounds three changes: year shift, measurement-protocol shift (single-face $\to$ multi-face) and class-space change (SX $\to$ GS, novel class); the source-centroid for GS is undefined on this fold and ProtoNet-C automatically degenerates to $\alpha = 0$ for that class. By contrast, LOEO-2018 and LOEO-2019 are year-shift-dominated; their source domains include 2025, so they are not strictly pure year-shift folds either.

Episode construction. Each evaluation episode samples a 3-way K-shot support set and 15 queries per class (45 in total) from the target fold. Support and query sets are **strictly disjoint at the fruit-ID level**: every fruit's face scans go entirely to one side, ruling out within-fruit leakage. Meta-training episodes follow the same fruit-disjoint rule on source domains; repetitions use independent random seeds.

3.3. Encoder

The encoder f_θ is a 1D CNN–MLP hybrid (input dimension 229; two 1D convolutional layers with 32 and 64 channels, kernel sizes 7 and 5, batch-normalisation and ReLU, followed by a global average pool; a 256-dimensional fully connected hidden layer with ReLU and dropout 0.3 (Srivastava, Hinton, Krizhevsky, Sutskever and Salakhutdinov, 2014); and a 128-dimensional ℓ_2-normalised embedding head). Meta-training is episodic: each episode samples a 3-way K-shot task from the source domains and minimises the cross-entropy loss of the query predictions with the Adam optimiser (Kingma and Ba, 2015, $lr = 10^{-3}$); training runs for up to 500 epochs of 100 episodes each, with the checkpoint best on the source-validation split kept for evaluation. Vanilla ProtoNet (Snell et al., 2017) uses the mean of K support embeddings as the class prototype and assigns each query to the nearest prototype. As a controlled ablation we additionally evaluate **ProtoNet-C**, which interpolates the per-episode prototype $\mathbf{c}_k^{\text{task}}$ with the source-domain class centroid $\mathbf{c}_k^{\text{src}}$ via $\hat{\mathbf{c}}_k = (1-\alpha)\,\mathbf{c}_k^{\text{task}} + \alpha\,\mathbf{c}_k^{\text{src}}$, $\alpha \in [0,1]$ ($\alpha = 0$ collapses to vanilla ProtoNet, $\alpha = 1$ to a nearest-source-centroid classifier; bias–variance intuition: small source–target gap favours $\alpha > 0$, large gap favours $\alpha = 0$ (Ben-David et al., 2010)). In the cross-year setting here, $\alpha = 0$ is best overall and is the headline baseline.

3.4. Baseline methods and the unified single-face query protocol

Unified single-face query protocol. To ensure that cross-method numbers are not contaminated by an evaluation-protocol effect, every method in the headline panel is evaluated under the same query rule: at inference time, each query fruit is represented by exactly one randomly drawn face spectrum (already trivially satisfied for 2018/2019 single-face fruits, and enforced on 2025 by sampling a single face uniformly at random per fruit). Multi-face mean strategies are reported only in the dedicated face-mean ablation (Section 3.6). All numbers in the main K-shot table (Section 4.4) follow this rule.

Four headline few-shot baselines are reported in Table 6. The **SVM-Kshot** baseline uses a C-support vector classifier (C-SVC, RBF kernel) with $C = 10$ and γ = "scale" (defaults selected by a small grid search on the source-validation split), refitted on the K target support samples at inference time. **PLS-DA-Kshot** builds a PLS projection from the K support samples, with the number of latent components set to $\min(K \cdot |C| - 1,\, 229,\, |C|)$ where $|C|$ is the number of classes (or 1 when $K \le 2$). **Vanilla ProtoNet** is the prototypical-network classifier on the fold-restricted encoder, reported across $K \in \{1, 3, 5, 10, 20, 50\}$ under single-face evaluation. **Nearest centroid** classifies each query in the input space (SNV-normalised) by the nearest support mean, without any meta-training. The exploratory **ProtoNet-C** ($\alpha = 0.4$) source-centroid-fusion variant is reported only in the controlled α-ablation (Section 4.6); it is not part of the headline panel.

1D CNN and 1D ResNet supervised closed-set sanity checks. In addition to the four few-shot baselines we report two modern deep baselines that do not rely on the K-shot framework. The **1D CNN** is a four-layer 1D convolutional classifier (filters 32/64/128/128, kernel sizes $7, 5, 5, 3$, two max-pool stages, global average pool, two-layer MLP head with ReLU and dropout 0.3, output dimension $|C|$ origins). The **1D ResNet** has a 7-kernel input convolution, max-pool, three residual blocks of two 3-kernel convolutions each (channel width 64), a global average pool and a linear classifier head. Both are trained with cross-entropy on every face spectrum in the source-fold years (2018+2025 for the LOEO-2019 fold, etc.; the target-fold year is excluded entirely from training), Adam at $lr = 10^{-3}$ with weight decay 10^{-4}, cosine annealing across 60 epochs, and batch size 128. Each query fruit's prediction at evaluation uses one randomly drawn face spectrum (single-face query rule); we average macro F_1 over five random seeds per fold. They are reported as supervised closed-set sanity checks rather than as a supervised, full-source extrapolation of the few-shot panel, because the closed-set softmax has no mechanism for the unseen Gansu class on LOEO-2025.

Classical unsupervised domain-adaptation methods such as transfer component analysis (TCA) (Pan and Yang, 2010) and subspace alignment (SA) (Fernando, Habrard, Sebban and Tuytelaars, 2013) are not included as baselines, because they require access to the full unlabelled target-domain data for distribution alignment. This is asymmetric with respect to the present episodic K-shot framework, in which only K labelled support samples are available at inference; a direct comparison would give those methods an unfair information advantage. A protocol-matched comparison—for example, a K-shot variant of TCA/SA—is left to future work and discussed further in Section 5.

3.5. Strict source-only z-score: pre-processing variant ablation

To make the pre-processing pipeline fully target-blind we additionally run the K-shot panel under a *strict source-only z-score* variant. For each LOEO fold, the per-band mean and standard deviation are computed only on the source-fold spectra and then applied identically to source and target. The variant is contrasted against the *weakly-transductive* year-wise z-score (each year normalises to its own per-band mean and standard deviation, including the target year's own statistics) and the *default* pipeline (SG smoothing followed by SNV with no z-score stage). The contrast quantifies how much the unlabelled per-band moment of the target year alone can move LOEO performance.

3.6. Face-mean ablation (LOEO-2025 fold only)

The 2025 subset provides about five measurement positions per apple, which lets us systematically test how multi-face aggregation affects classification. The face-mean ablation is reported separately from the headline cross-method comparison: it answers a different research question (how to allocate scanning budget between support side and query side) and is performed only on the LOEO-2025 fold, the single fold with multi-face data. We define a 2×2 design that crosses support-side strategy (single-face vs. multi-face mean) with query-side strategy (single-face vs. multi-face mean): $S_{\text{single}}Q_{\text{single}}$ (baseline), $S_{\text{multi}}Q_{\text{single}}$ (support-side mean only), $S_{\text{single}}Q_{\text{multi}}$ (query-side mean only), $S_{\text{multi}}Q_{\text{multi}}$ (both sides averaged). The ablation is performed at $K = 5$ over 50 repetitions. We also include an SVM-Kshot control, in which each face measurement of a multi-face apple is treated as an independent training sample, giving SVM access to about $5K$ effective training instances. None of the face-mean numbers feed into the main K-shot table (Section 4.4).

3.7. Multi-seed variance analysis design

Cross-paper comparability of an NIR benchmark requires the sources of evaluation uncertainty to be made explicit and individually quantified. We separate randomness into three layers: (i) *data-split layer*—LOEO-Year is deterministic at the year level and has no random-split degree of freedom; (ii) *encoder-training layer*—each LOEO fold's encoder is independently re-trained from scratch under five random seeds (42–46), with all other training hyper-parameters (architecture, Adam $lr = 10^{-3}$, CosineAnnealing, patience $= 12 \times$ 100 episodes, $n_{\text{way}} = 3$, $k_{\text{support}} = 5$, $q_{\text{query}} = 10$, temperature $= 10$) identical to those of Section 3.3; (iii) *episode-sampling layer*—each encoder is then evaluated over 50 independent episode seeds (42–91).

This yields a fully crossed design of **3 folds $\times$ 5 encoder seeds = 15 independent encoders**, each evaluated at **6 K values $\times$ 50 episode seeds $\times$ 4 methods (vanilla ProtoNet $\alpha = 0$, SVM, PLS-DA, nearest centroid)** $= 18{,}000$ **evaluations** in total. All four methods follow the unified single-face query protocol of Section 3.4, identical to the headline panel of Table 6. The SHA-256 hash (first 16 hex digits), source-validation accuracy and best-stopping episode of every encoder checkpoint are logged. A complete experiment bundle—encoder metadata, long-format per-episode F1 records, summary variance-decomposition tables and direction-consistency tables—will be deposited with the paper.

Evaluation variance is quantified along two axes: (a) the **between-encoder standard deviation**, computed by first averaging the 50 episode seeds within each encoder and then taking the standard deviation across the five encoders; (b) the **within-encoder standard deviation**, computed as the mean of per-encoder episode standard deviations. The ratio of the two reflects how much "re-training the encoder under a different seed" perturbs a result relative to "drawing

Table 2
Source-validation accuracy of the fold-restricted encoder on the three LOEO folds.

LOEO fold	Source domains	Val. accuracy
2018 held out	2019 + 2025	0.9478
2019 held out	2018 + 2025	0.9411
2025 held out	2018 + 2019	0.9478
Three-fold mean	—	0.9456

a different episode sample," and therefore delimits the regime in which the conventional "50-repetition standard deviation" suffices as a single uncertainty descriptor (see Section 4.2).

3.8. Statistical tests

The headline method comparison (vanilla ProtoNet vs. SVM/PLS-DA) in this manuscript is reported descriptively from three-fold mean macro F_1 values (Section 4.5). Per-seed paired statistics (paired t tests, Wilcoxon signed-rank tests (Wilcoxon, 1945), Cohen's d (Cohen, 1988), Benjamini–Hochberg correction (Benjamini and Hochberg, 1995)) are not used here because they would only quantify resampling stability under repeated episode draws on a fixed dataset (Bouckaert, 2004) rather than population-level inference across independent harvest seasons; the deposited episode logs include the per-seed values needed to compute them if a downstream user wants to. The directional conclusions in this paper rest on three-fold mean consistency only.

4. Results

4.1. Fold-restricted encoder validation

Before reporting LOEO performance, it is necessary to confirm the discriminative ability of each fold-restricted encoder on its source domain, because an under-trained encoder would interfere with the downstream comparison. The source-validation accuracies on the 2018, 2019, and 2025 folds are 0.9478, 0.9411, and 0.9478 respectively (Table 2), with a three-fold mean of 0.9456, which shows that the encoder cleanly separates origins within its training distribution. The slight drop on the 2019 fold (0.9411) is consistent with the larger spectral overlap between Shandong and Shanxi in that year's data. Strong within-domain discrimination does not, however, imply strong cross-year transfer; the question of interest is how much performance drops when the encoder is used for few-shot classification on the held-out target year.

4.2. Encoder-layer vs. episode-layer variance decomposition

Following the seed-matrix design of Section 3.7, the 15 fold–seed encoders re-trained from scratch reach source-validation accuracies in the range $[0.9256, 0.9611]$ (median 0.9478; full per-encoder metadata in Table 3), in tight agreement with the three-fold mean of 0.9456 reported in Table 2. Every single-seed result falls within this narrow band, indicating that the encoder-training procedure itself is highly stable. The full 18,000-evaluation panel takes about 90 minutes on Apple Silicon MPS.

Variance decomposition (vanilla ProtoNet, $\alpha = 0$, three-fold mean). Table 4 reports the between-encoder and within-encoder standard deviations across K. As K grows, the within-encoder (episode-layer) variance is suppressed by averaging more support samples, while the between-encoder (initialisation-layer) variance is essentially K-independent; the ratio narrows monotonically from $1:8.5$ at $K = 1$ to $1:1$ at $K = 20$–50. The implication is that the conventional 50-repetition standard deviation is an adequate single-number proxy in the small-K regime, but **at $K \geq 20$ roughly half of the residual uncertainty originates from encoder initialisation**—a component that single-seed reports systematically fail to quantify.

Encoder-initialisation sensitivity. The cross-encoder macro-F_1 range (per_enc_range, three-fold mean) is 0.041 at $K = 5$ and $\sim$0.05 at $K = 20$–50. Among the three folds, **LOEO-2025 is the most encoder-sensitive at large K (range up to 0.084–0.086)**, because the Gansu novel-class prototype is built entirely from target-side support samples and is therefore particularly sensitive to the geometry of the encoder embedding space. This per-fold asymmetry has methodological implications for any future extension of the benchmark to additional novel origins.

Direction stability of method selection. Of the 18 (fold $\times K$) combinations, 14 ($\sim$78%) show unanimous agreement among the five encoders on whether ProtoNet or SVM is preferred. The remaining four are fold–K *boundary cases* (LOEO-2018 $K = 5$, LOEO-2019 $K = 50$, LOEO-2025 $K = 10$ and $K = 20$) at which the cross-encoder ProtoNet–SVM gap is comparable in magnitude to the encoder-initialisation variance, so the five-seed vote splits 4:1 or 3:2 rather than 5 : 0; this is exactly the "unstable-near-the-decision-boundary" signal that the seed-matrix design is meant to expose. Away from these four boundary cases the method-selection conclusions of Table 6 are robust to the encoder training seed.

The fold-level crossover positions are, however, spread across the K axis: on **LOEO-2018** the crossover sits between $K = 1$ and $K = 3$ (SVM wins from $K \geq 3$); on **LOEO-2025** it sits between $K = 10$ and $K = 20$ (ProtoNet wins through $K \leq 10$); and **LOEO-2019 does not cross within the tested range**—vanilla ProtoNet leads SVM throughout $K \in \{1, \ldots, 50\}$, with the gap closing only at $K = 50$ to within the encoder-initialisation noise. The headline three-fold-mean crossover region of $K = 10$–20 in Table 6 is therefore an artefact of pooling these three

Table 3
Source-validation accuracy, training stop-episode and SHA-256 (first 16 hex digits) of every encoder checkpoint in the 15-encoder seed-matrix panel (3 folds $\times$ 5 seeds). All accuracies fall in $[0.9256, 0.9611]$, in line with the three-fold mean of 0.9456 reported in Table 2.

LOEO fold	Encoder seed	Val. acc	Stop ep.	SHA-256 (16)
2018	42	0.9478	1700	49b655ea642b7446
2018	43	0.9456	1900	83ba47556bc9c0ca
2018	44	0.9433	1300	bee31efbc3d3aec2
2018	45	0.9489	1900	b16f7f7050b7cf5b
2018	46	0.9467	1900	5718c7abcdc31259
2019	42	0.9478	1900	eb6d8daf7637f81c
2019	43	0.9356	1500	1abe7bba972b9f3f
2019	44	0.9367	1800	e920c13c002221fe
2019	45	0.9256	2000	a6861c8f74c69498
2019	46	0.9400	1800	5abaaea5ef686cb0
2025	42	0.9611	1800	d0965ec662865ace
2025	43	0.9467	2000	e3ad54835216bffb
2025	44	0.9567	1900	846197adf448a1f9
2025	45	0.9567	1700	31ef3644cffdad0e
2025	46	0.9489	2000	87e29720f7c30b51

different per-fold positions (one of which does not cross at all); **no individual fold actually crosses inside that region**, and practical method selection should be guided by the source-domain configuration most similar to the deployment scenario rather than by the global three-fold mean.

Consistency with the headline numbers. The five-encoder three-fold mean for vanilla ProtoNet at $K = 5$ is 0.6547, against 0.665 for the headline entry in Table 6 (difference $\Delta = -0.010$, well inside the total standard deviation of ± 0.049 at $K = 5$ in Table 4). The seed-matrix panel tracks the headline numbers for the other three baselines as well: at $K = 5$ the SVM three-fold mean is 0.606 (Table 6: 0.604, $\Delta = +0.002$), PLS-DA 0.625 (Table 6: 0.628, $\Delta = -0.003$), nearest centroid 0.613 (Table 6: 0.616, $\Delta = -0.003$); across all 24 (method $\times K$) entries of Table 6 the per-cell deviation never exceeds $|0.010|$. Per fold for vanilla ProtoNet: LOEO-2018 0.674 vs. Table 7 0.688 ($\Delta = -0.014$); LOEO-2019 0.711 vs. 0.724 ($\Delta = -0.013$); LOEO-2025 0.579 vs. 0.584 ($\Delta = -0.005$). The headline numbers therefore lie squarely inside the cross-encoder-seed distribution and are not driven by an unusually favourable seed, confirming the seed-robustness of the published baselines.

4.3. Evaluation contamination: shared encoder vs. fold-restricted encoder

As shown in Table 5, at $K = 5$ the shared encoder gives a three-fold mean macro F1 of 0.931, while the fold-restricted vanilla-ProtoNet baseline (single-face query) reaches only 0.665, with a gap of $\Delta = +0.266$. The exploratory ProtoNet-C ablation ($\alpha = 0.4$) sits even lower at 0.559, but its three-fold gap of $+0.372$ to the shared encoder reflects both encoder contamination

Table 4
Variance decomposition for vanilla ProtoNet ($\alpha = 0$) under 5 encoder seeds $\times$ 50 episode seeds (three-fold mean; macro F_1). "Between-enc. SD" is the standard deviation across the five per-encoder means; "within-enc. SD" is the mean of per-encoder episode standard deviations; "Enc. range" is the largest minus the smallest per-encoder mean.

K	3-fold mean	Between-enc. SD	Within-enc. SD	Ratio	Enc. range
1	0.583	0.009	0.077	1 : 8.5	0.023
3	0.627	0.016	0.060	1 : 3.7	0.041
5	0.655	0.017	0.047	1 : 2.8	0.041
10	0.672	0.019	0.028	1 : 1.5	0.043
20	0.678	0.021	0.021	1 : 1.0	0.050
50	0.684	0.020	0.019	1 : 1.0	0.049

and a source-centroid bias; a conservative estimate of pure encoder leakage uses the vanilla ProtoNet baseline as the reference ($\Delta = +0.266$). Evaluation contamination was most severe on the 2025 fold, where the shared and fold-restricted-vanilla encoders gave 0.917 and 0.584 respectively ($\Delta = +0.333$). The larger gap on the 2025 fold is consistent with the additional protocol mismatch in that fold: the 2025 data contain multi-face measurements that are absent from 2018 and 2019, so a shared encoder trained on all three years can learn multi-face-specific representations that simplify the 2025 episodes at evaluation.

The contamination gap reported here only holds under the specific experimental conditions of this study: four-origin apple NIR (Xinjiang, Shandong, Shanxi, Gansu), a single hand-held instrument (590–1001 nm), and the LOEO-Year protocol described above. Under different dataset structures, instrument configurations, or evaluation protocols, the magnitude of contamination may differ; the numerical value should be read as a reference under this benchmark, not as a universally applicable constant.

These results indicate that LOEO macro F_1 values reported above 0.9 under shared-encoder protocols (a common practice in the NIR meta-learning literature) substantially overestimate the actual cross-year generalisation ability, because the encoder has been exposed to target-year data before evaluation (Fig. 3). The fold-restricted encoder protocol is therefore the more appropriate evaluation standard for cross-year NIR benchmarks, and all subsequent results in this paper use fold-restricted encoders.

Table 5
Evaluation contamination at $K = 5$: shared encoder vs. fold-restricted encoders (Configuration C, $\alpha = 0$, the best fold-restricted configuration), means over 50 repetitions on the default SG + SNV pipeline (Variant O). The three-fold-mean gap between the shared encoder and Configuration C is $+0.266$. Configuration C is fully target-blind under this default pipeline; the strict source-only z-score variant (B) is reported separately in Section 4.7.

Protocol	2018	2019	2025	Mean
Shared encoder (contaminated)	0.939	0.937	0.917	0.931
Fold-restr. vanilla ProtoNet ($\alpha = 0$)	0.688	0.724	0.584	**0.665**
Fold-restr. ProtoNet-C ($\alpha = 0.4$, abl.)	0.684	0.584	0.409	0.559

Table 6
Macro F_1 of the four few-shot baselines at each K (three-fold mean; 50 repetitions per fold; **unified single-face query protocol**). Bold marks the best method at each K. ProtoNet-C ($\alpha = 0.4$) is reported only as the controlled ablation in Section 4.6; the 1D CNN and 1D ResNet supervised baselines (full-source training, not few-shot) are in Table 8. See Table 7 for the per-fold breakdown.

K	ProtoNet	SVM	PLS-DA	N.c.
1	**0.588**	0.521	0.477	0.543
3	**0.637**	0.570	0.594	0.592
5	**0.665**	0.604	0.628	0.616
10	**0.675**	0.656	0.658	0.634
20	0.685	**0.698**	0.678	0.645
50	0.690	**0.758**	0.692	0.656

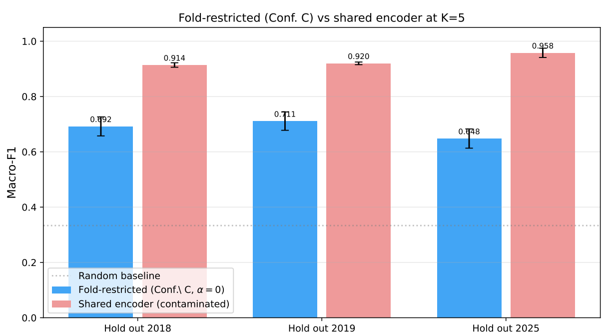

Figure 3. Macro F_1 at $K = 5$ for the fold-restricted encoder (re-trained per fold) and the shared encoder (trained on all years), on each LOEO-Year fold. The shared encoder is consistently higher across the three folds, with the largest gap on LOEO-2025.

4.4. Cross-method comparison across K-shot regimes

Result-interpretation framework. Following the fold taxonomy from the methods section, we report results in two layers: (i) *cleaner year-shift-dominated folds*—LOEO-2018 and LOEO-2019, whose target side is consistent in measurement protocol and class space with one of the other source years (note: their source domains do include 2025, so they are not strictly pure year-shift folds); (ii) *compound-shift stress test*—LOEO-2025, which simultaneously confounds year, protocol and class-space change. The three-fold mean is provided as a summary; cross-year-generalisation conclusions rely primarily on the cleaner year-shift-dominated folds, with LOEO-2025 interpreted separately. The following analyses and the discussion in Section 5 adhere strictly to this taxonomy. **All numbers in the headline cross-method panel use the unified single-face query protocol on every fold** (Section 3.4); multi-face strategies are reported separately in the face-mean ablation (Section 4.9).

The macro F_1 of the four few-shot baselines as a function of K is shown in Fig. 4 and Table 6; ProtoNet-C is reported in the controlled α-ablation (Section 4.6) and the supervised 1D CNN/ResNet sanity checks in Section 4.8.

In the small-to-moderate-sample regime ($K \leq 10$), the meta-trained encoder dominates: vanilla ProtoNet leads the three-fold mean at $K = 1$ (0.588), $K = 3$ (0.637), $K = 5$ (0.665) and $K = 10$ (0.675). At $K = 1$, vanilla ProtoNet is $+0.045$ F1 above nearest centroid and $+0.111$ F1 above PLS-DA. ProtoNet-C ($\alpha = 0.4$) is close behind vanilla ProtoNet on the LOEO-2018 fold but consistently lower on LOEO-2019 and LOEO-2025 because the source centroid pulls the prototype away from the target distribution. As K grows past 10, the SVM classifier overtakes the prototype-based methods: SVM reaches 0.698 at $K = 20$ and 0.758 at $K = 50$, while vanilla ProtoNet plateaus at 0.690 at $K = 50$ and ProtoNet-C plateaus lower at 0.566. The crossover between vanilla ProtoNet and SVM sits between $K = 10$ and $K = 20$; PLS-DA stays below vanilla ProtoNet through $K = 20$ (0.678 vs. 0.685) and only roughly ties at $K = 50$ (0.692 vs. 0.690). Below the crossover, the inductive bias accumulated through episodic meta-training compensates for an under-determined support set; above it, a fixed-capacity prototype is outperformed by classifiers with richer decision boundaries.

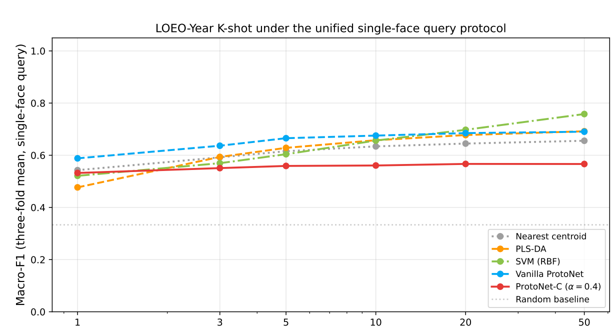

Table 7
Per-fold macro F_1 at $K = 5$ under the unified single-face query protocol (50 repetitions per fold; mean only; bold = best per fold). The LOEO-2025 fold confounds year, protocol and class-space change; results on this fold should be read as a compound-shift stress test rather than evidence for cross-year generalisation.

Fold	N.c.	PLS-DA	SVM	ProtoNet
LOEO-2018	0.675	**0.729**	0.681	0.688
LOEO-2019	0.642	0.624	0.632	**0.724**
LOEO-2025	0.529	0.532	0.499	**0.584**

N.c. = nearest centroid. All four methods evaluate every query fruit using exactly one randomly drawn face spectrum. ProtoNet-C ($\alpha = 0.4$) is reported only as the controlled ablation in Section 4.6.

Figure 4. LOEO macro F_1 of the four headline few-shot baselines as a function of K under the unified single-face query protocol (three-fold mean, 50 repetitions); the figure additionally plots the exploratory ProtoNet-C ($\alpha = 0.4$) curve as a dashed grey reference for the controlled ablation in Section 4.6. Vanilla ProtoNet leads through $K = 10$; SVM overtakes at $K = 20$ and $K = 50$, while PLS-DA stays below vanilla ProtoNet until roughly $K = 50$.

To support per-fold interpretation and avoid pooling year-shift-dominated folds with the compound-shift 2025 fold, Table 7 reports the same methods at $K = 5$ per fold under the unified single-face query protocol. On LOEO-2018, PLS-DA narrowly leads (0.729) over vanilla ProtoNet (0.688); on LOEO-2019, vanilla ProtoNet leads (0.724) clearly; on the compound-shift LOEO-2025 fold the four classes-space-aware methods collapse to the 0.50–0.59 range while ProtoNet-C ($\alpha = 0.4$) drops further to 0.409, reflecting the joint year/protocol/class-space shift on this fold. Pooling the three folds into the headline three-fold mean therefore mixes a per-fold winner pattern in which no single method dominates uniformly.

4.5. Cross-method comparison: descriptive analysis under the unified protocol

Scope. This subsection is descriptive. Three-fold means are reported in Table 6; the deposited episode logs contain the per-seed values needed for paired statistics, which are not reproduced here because pseudo-replication on a fixed dataset would inflate any nominal significance (Bouckaert, 2004) (Section 3.8).

Under the unified single-face protocol, vanilla ProtoNet leads the three-fold mean over SVM by $\Delta = +0.067$ ($K = 1$), $+0.067$ ($K = 3$), $+0.061$ ($K = 5$) and $+0.019$ ($K = 10$), then loses to SVM at $K = 20$ ($\Delta = -0.013$) and $K = 50$ ($\Delta = -0.068$). Against PLS-DA the gap is $\Delta = +0.111, +0.043, +0.037, +0.017, +0.007$ at $K = 1, 3, 5, 10, 20$ and ties at $K = 50$ ($\Delta = -0.002$). At the per-fold level the picture is non-uniform: at $K = 5$ vanilla ProtoNet wins on LOEO-2019 but PLS-DA narrowly wins on LOEO-2018; at $K = 50$ SVM wins on LOEO-2018 and LOEO-2025 while vanilla ProtoNet still wins on LOEO-2019 (0.744 vs. 0.734). The crossover region between $K = 10$ and $K = 20$ in the three-fold mean should therefore be read as a benchmark-specific stability statement, not as a population-level inference; we report descriptive three-fold means here and rely on per-fold consistency rather than nominal significance, in line with the reasoning in Section 3.8.

4.6. Source-prototype fusion-weight ablation (brief)

A continuous sweep over $\alpha \in [0, 1]$ at $K = 5$ (21 values, 50 repetitions per value, fold-restricted encoder) was originally run with the multi-face query protocol on LOEO-2025; the directional conclusion is the same under either query protocol because the source-centroid bias scales with α regardless. The sweep confirms that **$\alpha = 0$ (vanilla ProtoNet) is the best overall and a robust default**: strictly monotonically decreasing on LOEO-2019 and LOEO-2025, and effectively flat on LOEO-2018 (peak at $\alpha \approx 0.25$ with $F_1 = 0.694$, only $+0.006$ above $\alpha = 0$). At the representative non-zero weight $\alpha = 0.4$, the three-fold mean under the unified single-face protocol (the same protocol as Table 6) is 0.559 (0.684/0.584/0.409 per fold from the unified-protocol results), against 0.665 for $\alpha = 0$ (0.688/0.724/0.584). The takeaway is that under cross-year apple NIR the fold-restricted episode prototypes should be computed from target-year support samples alone; the source-centroid bias dominates whenever the source–target gap is non-trivial. All subsequent "ProtoNet-C" references in this paper denote the controlled ablation with $\alpha > 0$; the default and best-performing configuration is vanilla ProtoNet ($\alpha = 0$), reported as the headline baseline.

4.7. Pre-processing variant robustness check (fixed encoder)

Scope and caveat. This is a fixed-encoder robustness experiment, not a clean preprocessing-protocol comparison: the fold-restricted encoder used in all main results was meta-trained under the default SG + SNV pipeline (Variant O), so adding any z-score stage on top of it feeds the encoder an out-of-distribution input. With that caveat, three variants on the same cached encoder at $K = 5$ give three-fold mean macro F_1: Variant O (default SG + SNV, no z-score) 0.665 (0.688/0.724/0.584); Variant A (year-wise z-score using each year's own per-band statistics, weakly transductive) 0.524 (0.561/0.650/0.360); Variant B (strict source-only z-score, computed only on source-fold spectra) 0.559 (0.513/0.636/0.527). Per fold, Variant B is *lower* than Variant A on LOEO-2018 and LOEO-2019 but substantially *higher* on the compound-shift LOEO-2025 fold ($+0.167\,F_1$). The mean gap $\Delta_{B-A} = +0.035$

Table 8
1D CNN and 1D ResNet macro F_1 on the three LOEO-Year folds (five-seed mean, single-face query at evaluation, full source-fold training set). Class space is the full four-origin set {XJ, SD, SX, GS} so the LOEO-2025 fold has to predict GS without ever having seen it during training.

Method	LOEO-2018	LOEO-2019	LOEO-2025	3-fold
1D CNN	0.609	0.496	0.183	0.429
1D ResNet	0.555	0.475	0.191	0.407

is an order of magnitude smaller than the encoder-leakage gap of +0.266 from Section 4.3. When an explicit z-score is required for a cross-year benchmark, the strict source-only variant (B) is the clearly target-blind reference; the headline panel of this paper (Table 6) uses Variant O, which is also the encoder's training distribution. A clean comparison would retrain the encoder under each variant, which is left to future work.

4.8. 1D CNN and 1D ResNet supervised baselines

To complement the few-shot panel with modern deep baselines, we trained a 1D CNN and a 1D ResNet as supervised classifiers on the source-fold years only (target-fold year fully excluded from training), using cross-entropy on every face spectrum, Adam at $lr = 10^{-3}$ for 60 epochs, and evaluated on every target-fold fruit using one randomly drawn face spectrum (single-face query rule). Means and standard deviations are over five random seeds per fold (Table 8). They are reported separately from the headline few-shot panel as supervised closed-set sanity checks.

Caveat. The 1D CNN and 1D ResNet are closed-set softmax classifiers over the full {XJ, SD, SX, GS} class space; on the LOEO-2025 fold their training set contains no GS spectra at all, so the GS row of the softmax is essentially random at evaluation. They are therefore not a fair supervised, full-source comparison to the few-shot panel, which on the same fold builds a GS prototype from the K target support samples; we report them as supervised sanity-check baselines only. With this caveat, the deep baselines reach three-fold mean macro F_1 of 0.429 (CNN) and 0.407 (ResNet)—competitive with the K-shot panel on the LOEO-2018 fold, lagging on LOEO-2019, and collapsing on LOEO-2025 ($F_1 \approx 0.19$, near random for a four-class softmax that never saw the GS class). The few-shot panel's advantage on this benchmark therefore stems both from the meta-trained encoder's cross-domain exposure and from the architectural ability of the K-shot framework to handle a previously-unseen origin class.

Table 9
2×2 face-mean ablation on the LOEO-2025 fold ($K = 5$, 50 repetitions). Gains are reported relative to the $S_{\text{single}}Q_{\text{single}}$ baseline.

Strategy	Macro F1	Gain
$S_{\text{single}}Q_{\text{single}}$ (baseline)	0.584	—
$S_{\text{multi}}Q_{\text{single}}$ (support-side only)	0.595	$+0.011$
$S_{\text{single}}Q_{\text{multi}}$ (query-side only)	0.639	$+0.055$
$S_{\text{multi}}Q_{\text{multi}}$ (full mean, main)	**0.658**	$+0.074$
SVM-Kshot control	0.731	—

4.9. Decomposition of face-mean gain (LOEO-2025 fold ablation)

This is an exploratory case study restricted to the LOEO-2025 fold (the only year with multi-face data) at $K = 5$ over 50 repetitions; external validity to other folds and instruments remains to be verified. The 2×2 face-mean ablation (Table 9) decomposes the gain: support-side mean alone raises F_1 from 0.584 to 0.595 (+0.011, bootstrap 95% CI $[+0.001, +0.028]$); query-side mean alone raises it to 0.639 (+0.055, bootstrap 95% CI $[+0.040, +0.070]$); both sides reach 0.658, with a small positive interaction (+0.008). The dominant contribution thus comes from the query side, plausibly through (a) suppression of within-fruit biological noise and (b) test-time ensembling; the current design cannot separate these mechanisms. An SVM-Kshot control that treats each face as an independent training sample (effectively $5K$ instances) reaches 0.731, substantially exceeding all ProtoNet configurations—a non-protocol-matched comparison that nevertheless indicates the upper bound when face measurements are exploited as independent evidence.

4.10. Per-origin recognition difficulty

A per-class analysis under the fold-restricted encoder at $K = 5$ over 50 repetitions (full table in the deposited supplementary material) reveals a consistent pattern: **Xinjiang** is the easiest origin (recall 0.986/0.795/0.699 across the three folds), **Shandong/Shanxi** are the hardest among the established origins on LOEO-2018/2019 (recall 0.53–0.69, with substantial SD/SX spectral overlap), and **Gansu** on LOEO-2025 is a novel-class challenge (recall 0.533 with no source reference). A supplementary clean-fold check (source = only a single non-target year, excluding 2025) gives ProtoNet macro F_1 of 0.606 and 0.624 on LOEO-2018 and LOEO-2019, against 0.688 and 0.724 for the mixed source used in the main experiment; the mixed source consistently helps both ProtoNet and SVM, but on the clean folds SVM already matches or beats vanilla ProtoNet at $K = 5$ (2018: 0.620 vs. 0.606; 2019: 0.653 vs. 0.624), so the ProtoNet-vs-SVM crossover is *not* stable to source-domain composition. This is a benchmark-design caveat: the headline three-fold mean rests on the mixed-source configuration, and the crossover region is sensitive to which non-target years populate the source.

5. Discussion

5.1. Evaluation contamination: prevalence and remedies

The shared-encoder protocol produces substantial evaluation leakage ($\Delta F_1 = +0.266$ at $K = 5$, three-fold mean), which is a logical consequence of the workflow widely followed in the meta-learning community: the encoder is pre-trained on the full data including the test domain and then "frozen". In NIR specifically, the leakage has an additional amplification mechanism: a 229-dimensional input space with densely packed discriminative information lets the encoder learn protocol-specific structure (multi-face vs. single-face) that artificially lowers the difficulty of 2025 episodes. The leakage does not depend on the encoder architecture—fully connected, 1D CNN/ResNet or Transformer—as long as joint pre-training is used; the reported magnitude is specific to this dataset and the present 1D CNN–MLP encoder. A direct remedy is to retrain the encoder from scratch within each LOEO fold using source-domain data only, at a non-negligible computational cost. The takeaway for future cross-year NIR benchmarks is to specify the encoder protocol, treat fold-restricted performance as a necessary baseline, and label shared-encoder results as "upper-bound references" rather than primary claims.

5.2. K-shot crossover and deployment implications

The crossover between vanilla ProtoNet and SVM between $K = 10$ and $K = 20$ (Table 6) separates two deployment regimes; PLS-DA only roughly catches vanilla ProtoNet at $K = 50$. In the early-harvest-season scenario or first-time new-origin entry (very few labelled reference samples per origin), meta-trained ProtoNet has a substantial practical advantage. In a routine scenario with a moderate-scale reference set (e.g. port-of-entry inspection), SVM and PLS-DA have a continually rising performance curve and benefit from algorithmic simplicity and interpretability. There is no absolute superiority between the two classes; applicability depends on deployment conditions. The crossover is benchmark- and instrument-specific—this paper uses one hand-held NIR instrument and four Chinese origins, with wholesale-market labels rather than chain-of-custody-certified origin verification—and practitioners should re-estimate it under their own conditions and validate against certified samples.

5.3. Face-mean: surface heterogeneity as signal and noise

Within-fruit spectral heterogeneity (sun-exposed vs. shaded faces) is sometimes of the same magnitude as between-fruit differences (Lammertyn et al., 2000; Pissard et al., 2013). Our 2×2 ablation (Section 4.9) shows the gain from multi-position averaging is asymmetric: query-side averaging contributes $+0.055\,F_1$ and support-side averaging only $+0.011\,F_1$ on LOEO-2025 at $K = 5$. Practically, under a fixed scanning budget, concentrating multi-face scans on the test fruit (query side) is more cost-effective than on the reference set (support side); the conclusion is drawn from a single-fold ablation with limited statistical power on the support-side gain, and should be re-validated on other folds and instruments.

5.4. Compound-shift on LOEO-2025 and limitations

The LOEO-2025 fold confounds year shift, measurement-protocol shift (single-face $\rightarrow$ multi-face) and origin-class change (SX $\rightarrow$ GS, novel class); the independent contributions of the three factors cannot be decomposed under the present design, and an orthogonal three-factor design (year $\times$ protocol $\times$ origin) would be needed for a mechanistic attribution. The LOEO-2018 and LOEO-2019 folds offer relatively cleaner year-shift references; the contamination gap is also larger on 2025 (+0.333) than on 2018/2019 (+0.251 and +0.213), consistent with the shared encoder learning protocol-specific representations from the multi-face training data.

Sample sources and confounders. Samples were purchased from local wholesale markets, the cultivar is predominantly Fuji but not strictly controlled to a single sub-line, and harvest ripeness, storage conditions and supply-chain handling were not recorded; the NIR spectral differences can therefore encode the joint effects of origin, variety, ripeness, storage and circulation handling, and we restrict our traceability claim to origin-label discrimination under commercial-batch conditions. To isolate the pure geographic signal, future experiments would need a single sub-line, standardised harvest dates, controlled storage and systematically recorded metadata. **Geographic and instrument generalisability.** The dataset is limited to three to four Chinese origins and one (per-year) instrument; transfer to European PDO apples or to instruments covering different bands is an open question. **Methodological aspects.** We do not include classical unsupervised domain-adaptation baselines (TCA (Pan and Yang, 2010), SA (Fernando et al., 2013)) because they require unlabelled target data which is asymmetric with the few-shot framework here; a protocol-matched "few-shot" DA baseline is a valuable extension. The fold-restricted source-training paradigm advocated here is mechanistically closer to *domain generalisation* than *domain adaptation*, and meta-domain-generalisation baselines would be a natural complement. A logical next step is domain-gap-aware adaptive fusion, in which α is inferred dynamically from a source–target distribution distance rather than fixed.

6. Conclusion

This study contributes a nine-domain apple NIR benchmark covering three harvest years, four origins, 2,424 fruits and 4,529 spectra, together with a leakage-controlled LOEO-Year evaluation protocol and a quantitative analysis of evaluation contamination that has been overlooked in the existing literature. At $K = 5$, a shared encoder inflated the LOEO macro F_1 by +0.266 over the best fold-restricted scheme, and the gap widened to +0.333 on the 2025 fold. Fold-restricted encoder training and an explicit statement of which target-domain information was available during training should therefore be reported as required baselines in future LOEO benchmarks. The continuous $\alpha \in [0, 1]$ sweep showed that $\alpha = 0$ (fold-restricted episode prototypes, i.e. vanilla ProtoNet) is a robust overall default; LOEO-2018 alone admits a tiny non-zero peak near $\alpha \approx 0.25$ that is at the noise floor. In the three-fold mean of the mixed-source benchmark, vanilla ProtoNet leads at $K \in \{1, 3, 5, 10\}$ and SVM overtakes at $K = 20$ and $K = 50$, suggesting a benchmark-specific method-selection boundary indexed by available support size; per-fold the picture is non-uniform (PLS-DA narrowly leads on LOEO-2018 at $K = 5$, vanilla ProtoNet still wins on LOEO-2019 at $K = 50$), and the boundary is not stable to source-domain composition (Section 4.10). The 1D CNN and 1D ResNet are reported separately as supervised closed-set sanity checks and are *not* directly comparable to the few-shot panel because their closed-set softmax has no mechanism for the previously-unseen Gansu class on LOEO-2025. The face-mean ablation shows an asymmetric gain decomposition—query-side +0.055 versus support-side +0.011 in F_1—which is clear under the present experimental conditions but should be re-validated on additional folds and instruments. The dataset is held internally at Xinjiang Agricultural University and is available from the corresponding author on request, together with the evaluation code, to support direct reproduction and extension (the Data availability statement).

CRediT authorship contribution statement

Panlin Li: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Qingli Zhang**: Investigation, Validation, Formal analysis, Writing – review & editing. **Qitai Jiao**: Investigation, Resources, Data curation. **Jiamei Sun**: Investigation, Data curation, Validation. **Hua Huang**: Conceptualization, Supervision, Project administration, Resources, Funding acquisition, Writing – review & editing.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The apple near-infrared spectra used in this study are internal data of Xinjiang Agricultural University and have not been publicly released. Public deposit (e.g. to Zenodo, figshare or CSTR) is not authorised by the data owner at this time. Researchers who wish to reproduce or extend the experiments may request access from the corresponding author on a case-by-case basis, subject to the data-sharing policies of Xinjiang Agricultural University. For the journal's single-anonymized review process, an anonymous reviewer-only link to the prepared package can be arranged through the editorial office on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used a large-language-model assistant (Anthropic Claude) to perform English language editing and to suggest phrasing alternatives. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No generative AI or AI-assisted tool was used to generate, alter or fabricate experimental data, figures, or scientific conclusions.

Funding

This study was supported by the Science and Technology Innovation Team of the Xinjiang Uygur Autonomous Region (Tianshan Innovation Team) (2022TSYCTD0011), the National Natural Science Foundation of China (61367001), and the National Innovation Training Program for College Students 2025 (202510758022).

References

Barnes, R.J., Dhanoa, M.S., Lister, S.J., 1989. Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra. Applied Spectroscopy 43, 772–777. doi:10.1366/0003702894202832.

Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., Vaughan, J.W., 2010. A theory of learning from different domains. Machine Learning 79, 151–175. doi:10.1007/s10994-009-5152-4.

Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B 57, 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

Benmouna, B., Pourdarbani, R., Sabzi, S., Fernández-Beltrán, R., García-Mateos, G., Molina-Martínez, J.M., 2022. Convolutional neural networks for estimating the ripening state of Fuji apples using visible and near-infrared spectroscopy. Food and Bioprocess Technology 15, 2226–2236. doi:10.1007/s11947-022-02880-7.

Blanco, M., Villarroya, I., 2002. NIR spectroscopy: a rapid-response analytical tool. TrAC Trends in Analytical Chemistry 21, 240–250. doi:10.1016/S0165-9936(02)00404-1.

Bouckaert, R.R., 2004. Estimating replicability of classifier learning experiments, in: Proceedings of the 21st International Conference on Machine Learning (ICML), p. 15.

Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. doi:10.1023/A:1010933404324.

Bureau, S., Cozzolino, D., Clark, C.J., 2019. Contributions of Fourier-transform mid infrared (FT-MIR) spectroscopy to the study of fruit and vegetables: a review. Postharvest Biology and Technology 148, 1–14. doi:10.1016/j.postharvbio.2018.10.003.

Cen, H., He, Y., 2007. Theory and application of near infrared reflectance spectroscopy in determination of food quality. Trends in Food Science & Technology 18, 72–83. doi:10.1016/j.tifs.2006.09.003.

Chen, W.Y., Liu, Y.C., Kira, Z., Wang, Y.C.F., Huang, J.B., 2019. A closer look at few-shot classification, in: International Conference on Learning Representations, pp. 1–16.

Cohen, J., 1988. Statistical Power Analysis for the Behavioral Sciences. 2nd ed., Lawrence Erlbaum Associates, Hillsdale, NJ.

Cortes, C., Vapnik, V., 1995. Support-vector networks. Machine Learning 20, 273–297. doi:10.1007/BF00994018.

Danezis, G.P., Tsagkaris, A.S., Camin, F., Brusic, V., Georgiou, C.A., 2016. Food authentication: techniques, trends & emerging approaches. TrAC Trends in Analytical Chemistry 85, 123–132. doi:10.1016/j.trac.2016.02.026.

Eisenstecken, D., Stürz, B., Robatscher, P., Lozano, L., Zanella, A., Oberhuber, M., 2019. The potential of near infrared spectroscopy (NIRS) to trace apple origin: study on different cultivars and orchard elevations. Postharvest Biology and Technology 147, 123–131. doi:10.1016/j.postharvbio.2018.09.018.

Ekramirad, N., Khaled, A.Y., Doyle, L.E., Loeb, J.R., Donohue, K.D., Villanueva, R.T., Adedeji, A.A., 2022. Nondestructive detection of codling moth infestation in apples using pixel-based NIR hyperspectral imaging with machine learning and feature selection. Foods 11, 8. doi:10.3390/foods11010008.

Fearn, T., 2001. Standardisation and calibration transfer for near infrared instruments: a review. Journal of Near Infrared Spectroscopy 9, 229–244. doi:10.1255/jnirs.314.

Fernando, B., Habrard, A., Sebban, M., Tuytelaars, T., 2013. Unsupervised visual domain adaptation using subspace alignment, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 2960–2967. doi:10.1109/ICCV.2013.368.

Feudale, R.N., Woody, N.A., Tan, H., Myles, A.J., Brown, S.D., Ferre, J., 2002. Transfer of multivariate calibration models: a review. Chemometrics and Intelligent Laboratory Systems 64, 181–192. doi:10.1016/S0169-7439(02)00048-5.

Finn, C., Abbeel, P., Levine, S., 2017. Model-agnostic meta-learning for fast adaptation of deep networks, in: Proceedings of the 34th International Conference on Machine Learning, pp. 1126–1135.

Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., Lempitsky, V., 2016. Domain-adversarial training of neural networks. Journal of Machine Learning Research 17, 1–35.

Geladi, P., Kowalski, B.R., 1986. Partial least-squares regression: a tutorial. Analytica Chimica Acta 185, 1–17. doi:10.1016/0003-2670(86)80028-9.

Gonzalvez, A., Armenta, S., de la Guardia, M., 2009. Trace-element composition and stable-isotope ratio for discrimination of foods with protected designation of origin. TrAC Trends in Analytical Chemistry 28, 1295–1311. doi:10.1016/j.trac.2009.08.001.

He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778. doi:10.1109/CVPR.2016.90.

Hospedales, T.M., Antoniou, A., Micaelli, P., Storkey, A.J., 2022. Meta-learning in neural networks: a survey. IEEE Transactions on Pattern Analysis and Machine Intelligence 44, 5149–5169. doi:10.1109/TPAMI.2021.3079209.

Kingma, D.P., Ba, J., 2015. Adam: a method for stochastic optimization, in: Proceedings of the 3rd International Conference on Learning Representations (ICLR).

Lammertyn, J., Peirs, A., De Baerdemaeker, J., Nicolai, B., 2000. Light penetration properties of NIR radiation in fruit with respect to non-destructive quality assessment. Postharvest Biology and Technology 18, 121–132. doi:10.1016/S0925-5214(99)00071-X.

LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. Nature 521, 436–444. doi:10.1038/nature14539.

Li, J., Huang, W., Zhao, C., Zhang, B., 2013. A comparative study for the quantitative determination of soluble solids content, pH and firmness of pears by Vis/NIR spectroscopy. Journal of Food Engineering 116, 324–332. doi:10.1016/j.jfoodeng.2012.11.021.

Li, L., Li, B., Jiang, X., Liu, Y., 2022. A standard-free calibration transfer strategy for a discrimination model of apple origins based on near-infrared spectroscopy. Agriculture 12, 366. doi:10.3390/agriculture12030366.

Liu, B., Yu, X., Yu, A., Zhang, P., Wan, G., Wang, R., 2019. Deep few-shot learning for hyperspectral image classification. IEEE Transactions on Geoscience and Remote Sensing 57, 2290–2304. doi:10.1109/TGRS.2018.2872830.

Liu, Y., Zhang, Y., Jiang, X., Liu, Y., 2025. A feature-enhanced approach based on joint domain alignment and multi-order derivative spectral reconstruction for predicting apple firmness using Vis-NIR spectroscopy. Food Chemistry 476, 143457. doi:10.1016/j.foodchem.2025.143457.

Long, M., Cao, Y., Wang, J., Jordan, M.I., 2015. Learning transferable features with deep adaptation networks, in: Proceedings of the 32nd International Conference on Machine Learning, pp. 97–105.

Luo, W., Huan, S., Fu, H.Y., Wen, G., Cheng, H., Zhou, J., Wu, H., Shen, G., Yu, R.Q., 2011. Preliminary study on the application of near infrared spectroscopy and pattern recognition methods to classify different types of apple samples. Food Chemistry 128, 555–561. doi:10.1016/j.foodchem.2011.03.065.

Ma, T., Xia, Y., Inagaki, T., Tsuchikawa, S., 2018. Rapid and nondestructive evaluation of soluble solids content in apples by near-infrared hyperspectral imaging. Journal of Food Engineering 224, 53–61. doi:10.1016/j.jfoodeng.2017.12.028.

Mishra, P., Passos, D., 2021. Deep chemometrics: validation and transfer of a global deep near-infrared fruit model to use it on a new portable instrument. Journal of Chemometrics 35, e3367. doi:10.1002/cem.3367.

Mishra, P., Roger, J.M., Marini, F., Biancolillo, A., Rutledge, D.N., 2021. Chemometric pre-processing can negatively affect the performance of near-infrared spectroscopy models for fruit quality prediction. Talanta 229, 122303. doi:10.1016/j.talanta.2021.122303.

Nicolai, B.M., Beullens, K., Bobelyn, E., Peirs, A., Saeys, W., Theron, K.I., Lammertyn, J., 2007. Nondestructive measurement of fruit and vegetable quality by means of near infrared spectroscopy: a review. Postharvest Biology and Technology 46, 99–118. doi:10.1016/j.postharvbio.2007.06.024.

Pan, L., Wu, W., Hu, Z., Li, H., Zhang, M., Zhao, J., 2024. Updating apple Vis-NIR spectral ripeness classification model based on deep learning and multi-seasonal database. Biosystems Engineering 245, 164–176. doi:10.1016/j.biosystemseng.2024.07.010.

Pan, S.J., Yang, Q., 2010. A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering 22, 1345–1359. doi:10.1109/TKDE.2009.191.

Peng, X., Bai, Q., Xia, X., Huang, Z., Saenko, K., Wang, B., 2019. Moment matching for multi-source domain adaptation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1406–1415. doi:10.1109/ICCV.2019.00149.

Pissard, A., Fernández Pierna, J.A., Baeten, V., Sinnaeve, G., Lognay, G., Mouteau, A., Dupont, P., Rondia, A., Lateur, M., 2013. Non-destructive measurement of vitamin C, total polyphenol and sugar content in apples using near-infrared spectroscopy. Journal of the Science of Food and Agriculture 93, 238–244. doi:10.1002/jsfa.5779.

Rinnan, Å., van den Berg, F., Engelsen, S.B., 2009. Review of the most common pre-processing techniques for near-infrared spectra. TrAC Trends in Analytical Chemistry 28, 1201–1222. doi:10.1016/j.trac.2009.07.007.

Savitzky, A., Golay, M.J.E., 1964. Smoothing and differentiation of data by simplified least squares procedures. Analytical Chemistry 36, 1627–1639. doi:10.1021/ac60214a047.

Snell, J., Swersky, K., Zemel, R.S., 2017. Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems, pp. 4077–4087.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. Journal of Machine Learning Research 15, 1929–1958.

Sung, F., Yang, Y., Zhang, L., Xiang, T., Torr, P.H.S., Hospedales, T.M., 2018. Learning to compare: relation network for few-shot learning, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1199–1208. doi:10.1109/CVPR.2018.00131.

Tian, Y., Wang, Y., Krishnan, D., Tenenbaum, J.B., Isola, P., 2020. Rethinking few-shot image classification: a good embedding is all you need? arXiv preprint arXiv:2003.11539 .

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems, pp. 5998–6008.

Vinyals, O., Blundell, C., Lillicrap, T., Wierstra, D., Kavukcuoglu, K., 2016. Matching networks for one shot learning, in: Advances in Neural Information Processing Systems, pp. 3630–3638.

Walsh, K.B., Blasco, J., Zude-Sasse, M., Sun, X., 2020. Visible-NIR 'point' spectroscopy in postharvest fruit and vegetable assessment: the science behind three decades of commercial use. Postharvest Biology and Technology 168, 111246. doi:10.1016/j.postharvbio.2020.111246.

Wang, Y., Yao, Q., Kwok, J.T., Ni, L.M., 2020. Generalizing from a few examples: a survey on few-shot learning. ACM Computing Surveys 53, 63:1–63:34. doi:10.1145/3386252.

Westad, F., Marini, F., 2015. Validation of chemometric models – a tutorial. Analytica Chimica Acta 893, 14–24. doi:10.1016/j.aca.2015.06.056.

Wilcoxon, F., 1945. Individual comparisons by ranking methods. Biometrics Bulletin 1, 80–83. doi:10.2307/3001968.

Wold, S., Sjöström, M., Eriksson, L., 2001. PLS-regression: a basic tool of chemometrics. Chemometrics and Intelligent Laboratory Systems 58, 109–130. doi:10.1016/S0169-7439(01)00155-1.

Woodcock, T., Downey, G., Kelly, J.D., O'Donnell, C., 2007. Geographical classification of honey samples by near-infrared spectroscopy: a feasibility study. Journal of Agricultural and Food Chemistry 55, 9128–9134. doi:10.1021/jf071566b.